FrostChain E-Skin Crate — System Electronics Schematic

1 OFF-GRID POWER solar → LiFePO4 → 12 V / 5 V

PV1 30 W flex

U1 MPPT
PV+ BAT+
charger
GND 4S LiFePO4
CV 14.4 V

BT1 4S BMS
CHG+ P+
GND 12.8 V 6 Ah
LiFePO4 + BMS

+12V → F1 10 A

U2 LM2596-5.0
VIN OUT
ON/OFF
GND
FB

+12V → C1 680 µF
L1 33 µH
D1 SS34
C2 220 µF
+5V

2 SENSE graphite E-skin half-bridge + ratiometric sigma-delta front end

+3V3 → R1 active 100 kΩ
+3V3 → **U7 SHT31** (VDD, GND, SCL, SDA)
R7/R8 4.7 kΩ
BRIDGE_MID
R2 dummy 100 kΩ
+3V3 → R5 ref 100 kΩ 0.1% → TH1 NTC 100 kΩ
NTC_IN
INT
SD_FB (1-bit)
R6 fb 100 kΩ 0.1%
C3 10 nF C0G
BRIDGE_LOW (switched)
R3 100 Ω
Q1 2N7002
R4 100 k
EN_BRIDGE (20 ms / 1 s)

U3 Sipeed Tang Nano 9K

Gowin GW1NR-9 FPGA - 3 DECIDE

+5V → 5V
+3V3 → 3V3 out (VCCIO)
GND

Phase / chop sequencer (EN 20 ms)
Σ-Δ ratiometric ADC (LVDS comparator)
CIC³ + MA-5 (50 Hz null)
INT8 TCN (1 MAC, ~1.7 k params)
CUSUM + consensus gate

I2C_SCL
I2C_SDA
LVDS_A+
LVDS_B+
LVDS_A−
LVDS_B−
SD_FB
EN_BRIDGE

UART_TX
UART_RX
TAMPER_IRQ
PELTIER_REQ

Ratiometric: bridge, NTC and the 1-bit feedback GPIO all share VCCIO → supply drift cancels, no reference IC needed.
Excitation only 20 ms per 1 s (2 % duty, about 5 µW average).
Targets: ≥ 13 ENOB @ 10 Hz, INT8 within 1 % of FP32, latency ≤ 2 s, ≤ 1 false alarm / 1000 device-hours.

4 COOL Peltier lid driver (backup top-up only)

PCM packs hold 4 ± 2 °C for ≥ 8 h
TEC fires only if T_int → 5 °C
or the TCN predicts hold-time loss
PELTIER_REQ (FPGA) → ESP32 → PWM
Hot-side NTC → ESP32 ADC: thermal
cut-off + fan-stall lockout

U8 BTS7960 module (IBT-2)
RPWM
43 A half-bridge, duty-limited PWM, C5 1000 µF B+/B−
B+ M+ M− B−
EN VCC LPWM GND
+12V
+5V
TEC1-12706 12 V, 6 A max

PELTIER_PWM

5 SETTLE ESP32-S3 + 5G/GNSS + smart contract

UART

U4 ESP32-S3
RXD TXD GPIO_IRQ GPIO_REQ
TX1 RX1 MDM_PWR
5V GND
+5V

U5 Modem
5G / LTE + GNSS (SIM7600 class)
RXD TXD PWRKEY
MAIN GPS
5G/LTE GNSS
VBAT GND

ECDSA-signed events + GPS
→ Solidity quality-score contract
Score = 100 − Σ(T_act − T_set)·t − penalty
Payout: > 90 : 100 % | 70-90 : 80 %
≤ 70 : 50 % | tamper flag → review

+12V → **U6 12 V → 4.0 V** (VIN, VOUT, GND)
buck, 3 A (modem VBAT)
C6 1000 µF

12 V / PV | 5 V | 3V3 ratiometric rail | analog sense | digital / PWM | UART

R, C, L symbols per IEC/ANSI mix; values are design targets for Phase 1-2 SPICE.